

Random Prefix Perturbations Diversify Reasoning Trajectories with a Sharp Non-Monotonic Optimum in Small Language Models

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Abstract

Prepending random embedding-scale tokens to the input of a 4-bit quantized language model (Qwen3-4B) produces a sharp, non-monotonic improvement in chain-of-thought arithmetic reasoning. Two random tokens raise mean accuracy from 32% to 51.6% (+19.6 percentage points) across ten independent directions, while one token yields 42.7% and three or eight tokens revert to $\sim 44\%$. A first-token logit probe confirms that perturbation does not activate reasoning mode (which is already saturated at $>99.99\%$ under thinking-enabled decoding); instead, it modulates the downstream reasoning trajectory. Different perturbation directions solve *different task subsets*: ten two-token directions collectively cover 100% of tasks (25/25), compared to 80% for three-token and 92% for eight-token perturbations. The three-token oracle set is a strict subset of the two-token oracle, indicating that over-perturbation contracts the reachable task space. Cross-model validation confirms the effect on four models including Qwen3-8B at 8-bit (+12.8pp at $n=10$, oracle 80%, McNemar 16/0), DeepSeek-R1-Distill-1.5B (oracle 100%, McNemar 6/0), and phi-2 (+6.7pp, out-of-family). A within-model quantization control reveals that weight precision modulates perturbation sensitivity: Qwen3-8B at 4-bit yields a null result while 8-bit yields +16pp on the same tasks. DeepSeek exhibits task-selective recoverability: every baseline failure is rescued by at least one perturbation direction, though the average random direction is slightly harmful (-1.6 pp at $n=10$), establishing a distinction between oracle accessibility and mean-effect improvement. A dose-response scout on DeepSeek confirms the non-monotonic window generalizes: 1-token perturbation hurts (-12 pp), 2-token is the local optimum, and 3-token enters a direction-sensitive bifurcated regime. A grading-confound analysis reveals that the accuracy gains are primarily a *convergence* effect: the model can already compute correct answers 80% of the time (answer-anywhere), but perturbation improves the rate at which the correct answer appears as the final output ($32\% \rightarrow 51.6\%$). Cross-domain validation on five complex planning tasks reveals two additional phenomena: (1) perturbation rescues catastrophic attention-sink-induced generation failures (14-word baseline collapse $\rightarrow$ 650+ word complete plans across all five perturbation seeds), and (2) *evolved* soft prompts (via a trained latent scorer + evolutionary search) surface qualitatively different reasoning—security frameworks, architectural patterns, and diagnostic strategies absent from both baseline and random perturbation outputs. These results establish embedding-space perturbation as a training-free intervention that diversifies reasoning trajectories, with transfer dependent on reasoning substrate and quantization fidelity, and suggest that directed perturbation via evolution can access latent knowledge the model cannot reach under standard greedy decoding.

1 Introduction

Large language models possess reasoning capabilities that are not uniformly accessible through standard prompting. Recent work has shown that semantically meaningless prefix tokens—repeated punctuation [1], pause tokens [2], and trailing whitespace [3]—can measurably alter model behavior, sometimes improving reasoning performance.

We study a related but distinct regime: *continuous* perturbation in embedding space. Rather than prepending discrete tokens from the vocabulary, we inject random vectors at the model’s native embedding scale as soft prompt prefixes. This eliminates semantic content entirely, isolating the effect of perturbation energy and position from token identity.

Our key findings on Qwen3-4B (4-bit quantized) with 25 nested arithmetic tasks are:

1. **Sharp non-monotonic optimum.** Two random prefix tokens raise mean last-integer accuracy from 32% to 51.6% across ten directions (primarily a convergence effect; answer-anywhere accuracy is 80% at baseline); one token yields 42.7%, and three or eight tokens revert to $\sim 44\%$.
2. **Complete oracle coverage.** Ten two-token directions collectively solve 100% of tasks (25/25), compared to 80% (20/25) for three-token and 92% (23/25) for eight-token perturbations. The three-token oracle set is a strict subset of the two-token oracle: five tasks solvable at two tokens become frozen at three. At matched budget ($k = 3$), two tokens achieve 77% oracle coverage versus 60% for three tokens and 68% for eight tokens.
3. **Heterogeneous task selectivity.** Different perturbation directions solve different task subsets, but per-direction solve counts vary normally (std = 1.87, $p = 0.66$ vs. heterogeneous iid null). The diversity is in *which* tasks are solved, not in *how many*.
4. **Trajectory modulation, not mode gating.** A first-token logit probe shows that `<think>` probability is saturated ($>99.99\%$) under all conditions including baseline. Perturbation does not activate reasoning mode; it modulates the downstream reasoning trajectory within an already-active think mode.
5. **Cross-domain: attention sink avoidance.** On complex planning tasks, greedy baseline can fail catastrophically (14 words before collapsing into degenerate output). Two-token perturbation rescues these failures into complete 650+ word plans across all five seeds, demonstrating that soft prompt perturbation breaks attention-sink-induced degenerate generation paths.
6. **Cross-domain: latent knowledge access via evolution.** Evolved soft prompts (trained scorer + evolutionary search) surface qualitatively different reasoning—security frameworks (MITRE ATT&CK, HSM-backed credential rotation), architectural patterns (honeypot deployment, DMZ isolation), and diagnostic strategies (app-level idempotency analysis) absent from both baseline and random perturbation outputs. This suggests embedding-space perturbation can access latent knowledge the model possesses but cannot reach under standard decoding.

All experiments use greedy decoding (temperature = 0), making the observed variability across perturbation directions a form of *deterministic chaos* rather than sampling noise.

2 Related Work

Discrete prefix perturbations. Shi et al. [1] demonstrate that repeated punctuation tokens (e.g., “/” and “?”) improve reasoning accuracy by 1–5% across multiple models, attributing the effect to MLP activation redistribution. Our work studies the continuous embedding-space regime, finding substantially larger effects (+19.6pp mean at the optimum), direction independence, and task-selective redistribution—phenomena not observed in discrete token space. In a head-to-head comparison on our task set (Section 4.9), discrete “/” and “?” tokens at 2 tokens yield +4pp and +16pp respectively (mean +10pp), while continuous random noise yields +19.6pp—a 9.6pp advantage for continuous perturbation.

Pause tokens. Goyal et al. [2] show that training with learned pause tokens improves downstream performance. Our effect requires no training: random embeddings at inference time suffice.

Extra-token expressivity. London & Nagarajan [4] prove that additional tokens increase transformer expressivity. Our results provide an empirical instantiation: two tokens at the correct energy scale unlock reasoning that the model cannot access with standard prompting.

Deterministic chaos in transformers. Li et al. [5] introduce quasi-Lyapunov exponents for language models, establishing that small perturbations to hidden states grow exponentially through transformer layers. Our invariance-length analysis (Section 4.5) provides a complementary behavioral measure of this phenomenon under greedy decoding.

Attention sinks. Xiao et al. [8] document the attention sink phenomenon: early tokens accumulate disproportionate attention from all subsequent tokens regardless of semantic content. Our cross-domain planning results (Section 4.11) provide a practical consequence: when attention sink positions contain only standard prompt tokens, greedy decoding can lock into degenerate generation paths. Soft prompt perturbation in these positions disrupts the degenerate pattern, rescuing catastrophic failures.

Stochastic resonance. The non-monotonic dose-response is reminiscent of stochastic resonance [6], where noise improves signal detection in nonlinear systems. However, our statistical tests do not confirm the formal inverted-U relationship at the task level (Section 6), so we use the more conservative framing of “non-monotonic optimum.”

3 Methods

3.1 Model and Inference

We use Qwen3-4B [7] with 4-bit GPTQ quantization, running on a single NVIDIA RTX 5090 Laptop GPU (24GB VRAM). All generation uses greedy decoding (`temperature = 0`, `do_sample = False`) with `max_new_tokens = 1024` and thinking mode enabled.

3.2 Task Suite

We use 25 nested arithmetic expressions at “sweet-spot” difficulty (3-step expressions with intermediate results in the hundreds to thousands range, 32% baseline accuracy). Tasks are deterministically generated and fixed across all conditions.

3.3 Perturbation Protocol

We inject k random embedding vectors as soft prompt tokens before the user input. Each vector is sampled from $\mathcal{N}(0, I)$ and scaled to match the model’s embedding table RMS norm ($\sigma_{\text{emb}} \approx 0.022$), calibrated via `auto_calibrate()` over 1000 vocabulary samples.

3.4 Conditions

Condition	Tokens	Description
Baseline	0	No prefix tokens
Zero embedding	8	All-zero vectors
Mean embedding	8	Mean of embedding table
Force-think	0	Forced <code><think></code> token, no noise
Random noise	1, 2, 3, 8	Random $\mathcal{N}(0, I)$, RMS-matched
Discrete tokens	2	Repeated “/” and “?” tokens (Shi comparison)

Table 1: Experimental conditions. All use greedy decoding.

3.5 Scoring and Data Integrity

Correctness is determined by `verify_answer()`: the last integer in the generated response must exactly match the expected answer. All experiments with the current codebase store the extracted answer alongside the full response (up to 2000 characters) for independent verification.

4 Results

4.1 Non-Monotonic Dose-Response

Tokens	Accuracy	Std	n	Δ vs baseline
0	32.0%	—	1	—
1	42.7%	1.9%	3	+10.7pp
2	51.6%	7.9%	10	+19.6pp
3	44.0%	5.3%	10	+12.0pp
8	44.4%	7.0%	10	+12.4pp

Table 2: Dose-response: accuracy vs. number of random prefix tokens (n = number of independent directions tested). Two tokens is the sharp optimum. Three and eight tokens revert to $\sim 44\%$.

experiments/fig_dose_response.pdf

Figure 1: Non-monotonic dose-response curve. Error bars show standard error of the mean across n perturbation directions.

The relationship between token count and accuracy is sharply non-monotonic. Two random tokens produce the highest mean accuracy ($51.6\% \pm 7.9\%$ across ten independent directions). Adding a third token drops accuracy to $\sim 44\%$, matching the eight-token level. The three-token oracle set (20/25) is a strict subset of the two-token oracle (25/25), indicating that over-perturbation contracts the reachable task space rather than merely redistributing it.

4.2 Explicit Think-Prefix Control

Condition	Accuracy	Δ vs baseline
Baseline	32.0%	—
Explicit <think> prefix	40.0%	+8pp
2-tok random noise (mean, $n = 10$)	51.6%	+19.6pp

Table 3: Explicit think-prefix control. Teacher-forcing the first token to <think> adds +8pp. Random noise adds a further +11.6pp beyond the explicit prefix (mean across 10 directions). Note: a first-token logit probe (Section 4.7) shows that <think> probability is already >99.99% at baseline under thinking-enabled decoding, so the +8pp is not mode activation but a trajectory-priming effect of the explicit prefix.

A first-token logit probe (Section 4.7) confirms that <think> is the dominant first token under all conditions (probability >99.99%), including unperturbed baseline. The observed accuracy differences therefore reflect trajectory modulation within an already-active reasoning mode, not mode-switching. Teacher-forcing <think> as an explicit prefix adds +8pp; random noise provides an additional +11.6pp beyond this, indicating that the perturbation effect operates on the reasoning trajectory rather than on mode entry.

4.3 Oracle Efficiency

Because different perturbation directions solve different task subsets, running multiple perturbations and taking the oracle (any-correct) provides coverage well above individual accuracy. Figure 2 shows coverage as a function of budget k for each condition.

The non-monotonic optimum extends beyond per-direction accuracy to coverage efficiency: five tasks solvable at two tokens become frozen at three tokens, contracting the reachable task set rather than merely redistributing it. Two-token perturbation alone covers all 25 tasks; combined oracle across all conditions is therefore $25/25 = 100\%$.

Oracle coverage grows more slowly than the independence null (positive task correlation, $z = -1.35$ for 2-tok vs. $z = -8.25$ for 8-tok), indicating that latents fail on correlated hard tasks. The two-token condition has the least redundancy, closest to independence.

4.4 Task Selectivity Without Equalization

An initial observation at $n = 3$ two-token directions suggested solve-count equalization (all directions solving exactly 15/25 tasks, $p = 0.031$ vs. heterogeneous iid null). This did not replicate: at $n = 10$ directions, solve counts range from 9 to 15 (std = 1.87, $p = 0.66$), fully consistent with independent sampling at heterogeneous per-task solve rates. The initial pattern was small-sample noise. The robust finding is *task selectivity*: directions solve different subsets, enabling oracle gains, but the total number solved per direction varies normally.

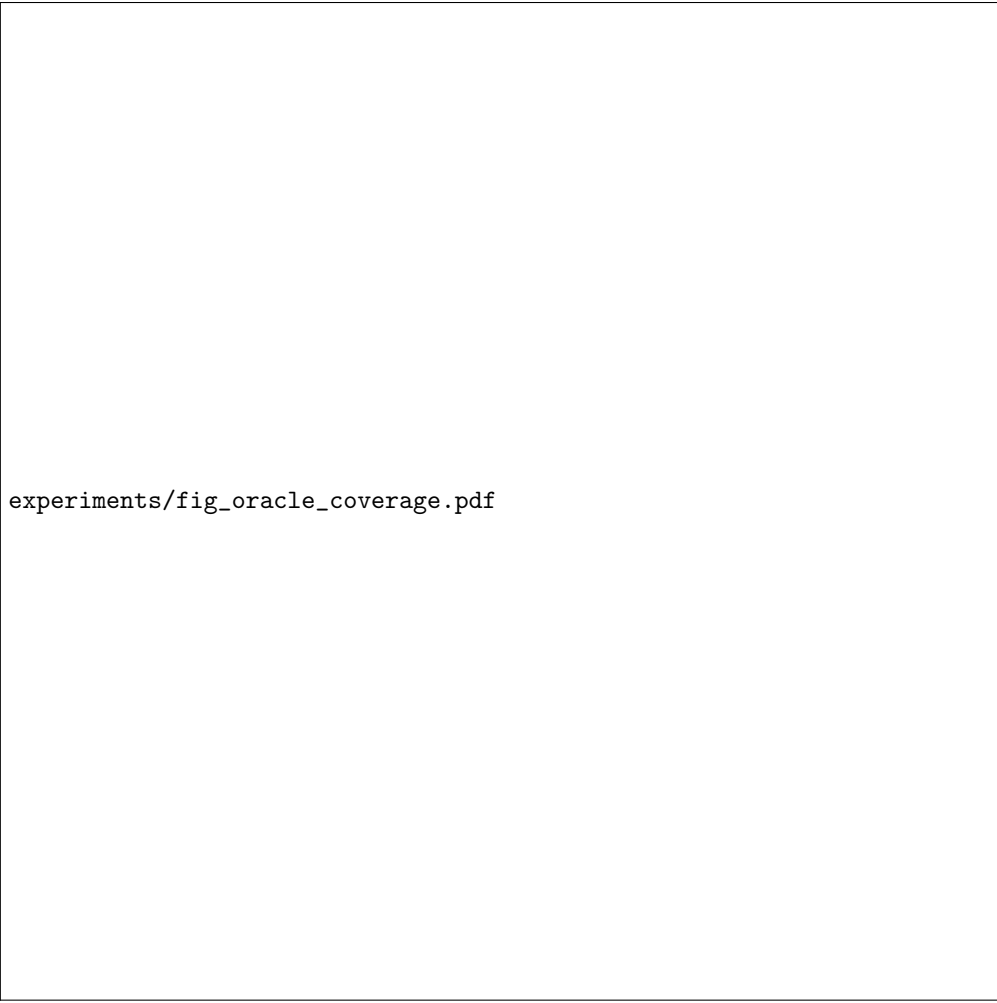
We test equalization by Monte Carlo simulation: given the observed per-task solve rates, we simulate 100,000 draws under a heterogeneous iid Bernoulli null (each task independently solved at its observed rate) and compare the observed standard deviation of per-direction solve counts against this null distribution.

We decompose tasks at each perturbation level into three categories: *unanimous* (solved by all sampled directions), *frozen* (solved by none), and *sensitive* (solved by some but not all). These categories are probabilistic, not absolute—additional directions may solve “frozen” tasks or fail “unanimous” ones.

4.5 Deterministic Chaos

Under greedy decoding, different random prefix vectors produce byte-identical text for an “invariance length” that decreases with perturbation energy:

The two-token sweet spot balances long enough invariance to preserve reasoning quality with short enough invariance for task-subset diversity.



experiments/fig_oracle_coverage.pdf

Figure 2: Oracle coverage vs. perturbation budget. Two-token (green) reaches 100% coverage at $k = 10$ (zero frozen tasks), dominates at every matched budget, and its oracle set is a strict superset of the three-token set (red, 80% at $k = 10$). Eight-token (blue) reaches 92% at $k = 10$.

4.6 Controls

- **Zero embedding** (+4pp): Sequence extension alone provides minimal benefit.
- **Mean embedding** (+4pp): Identical to zero—no advantage from the “average token.”
- **No-think mode**: When thinking is disabled (`enable_thinking=False`), perturbation has no effect. Chain-of-thought is the mediating mechanism.
- **Timing confound resolved**: The highest-accuracy 2-tok directions achieve 60% (15/25) at baseline-identical mean timing (73.6s), ruling out extra computation as the mechanism.

4.7 First-Token Logit Probe

To test whether perturbation activates reasoning mode, we extract logits at the first decode position (immediately after the assistant turn) under all conditions. The `<think>` token is rank 1 with probability >99.99% for all 25 tasks under every condition, including unperturbed baseline (Table 6).

This rules out first-token mode gating as the mechanism. The perturbation effect operates within an already-active reasoning mode, modulating the trajectory of the chain-of-thought rather than determining

Panel A: Matched budget ($k = 3$ directions)

Condition	Unan.	Frozen	Sens.	Mean acc.
1-tok	6.0	8.0	11.0	10.7/25
2-tok[†]	7.6	5.8	11.6	12.9/25
3-tok [†]	7.3	9.9	7.8	11.0/25
8-tok [†]	6.0	7.9	11.2	11.1/25

Panel B: Full observed directions (with iid p -values)

Condition	k	Unan.	Frozen	Sens.	Mean/25	Std	p_{iid}
1-tok	3	6	8	11	10.7	0.47	0.23
2-tok	10	6	0	19	12.9	1.87	0.659
3-tok	10	5	5	15	11.0	1.33	0.335
8-tok	10	3	2	20	11.1	1.76	0.586

Table 4: Task decomposition by perturbation magnitude. **Panel A:** At matched budget ($k = 3$), two tokens minimize the frozen set and maximize per-direction accuracy. [†]Values are averaged over all $\binom{n}{3}$ subsets (120 combinations each for $n = 10$ conditions). **Panel B:** Full observed directions with Monte Carlo p -values against heterogeneous iid null. No condition shows significant variance suppression; the diversity is in task identity, not solve count.

Tokens	Mean Invariance	Oracle
1	349 chars	68%
2	323 chars	100%
8	93 chars	92%

Table 5: Invariance length (byte-identical prefix before divergence) under greedy decoding. This is deterministic chaos, not sampling noise.

whether the model reasons at all.

4.8 Task-Specific Resonance Windows

Individual tasks exhibit condition-dependent resonance windows:

- **nest_005** (answer = 8360): solvable at 1-tok and 2-tok but frozen at 3-tok and 8-tok
- **nest_008** (answer = 7278): solvable *only* at 2-tok (the sole task uniquely rescued by a single condition)
- **nest_021** (answer = 5639): solvable at 2-tok and 8-tok but frozen at 1-tok and 3-tok

More noise can *prevent* solving certain tasks while enabling others. At $n = 10$ directions, the 2-tok condition has zero frozen tasks, while 3-tok freezes 5 and 8-tok freezes 2.

4.9 Shi et al. Comparison

To compare continuous and discrete perturbation regimes, we replicate the Shi et al. [1] protocol on our task set: repeating discrete tokens “/” and “?” as 2-token prefixes, using native embeddings ($RMS \approx 0.020$, close to our target 0.022).

Both discrete tokens improve over baseline, consistent with Shi et al.’s observation that punctuation prefixes can aid reasoning. However, continuous random noise in embedding space produces a substantially larger effect (+19.6pp vs. +10pp mean discrete). The gap suggests that the continuous embedding manifold offers a richer space of perturbations than the discrete token vocabulary. Notably, “?” (+16pp) outperforms

Condition	Mean $P(\langle \text{think} \rangle)$	Rank = 1
Baseline	99.994%	25/25
Zero (2-tok)	99.994%	25/25
Mean (2-tok)	99.994%	25/25
Noise 1-tok	99.996%	25/25
Noise 2-tok	99.997%	25/25
Noise 3-tok	99.997%	25/25
Noise 8-tok	99.995%	25/25
Discrete / (2-tok)	99.998%	25/25
Discrete ? (2-tok)	99.997%	25/25

Table 6: First-token logit probe. $\langle \text{think} \rangle$ probability is saturated under all conditions including baseline. Perturbation does not activate reasoning mode; it modulates the subsequent reasoning trajectory.

Condition	Accuracy	Δ vs. baseline	n
Baseline (no prefix)	32.0%	—	1
Discrete “/” (2 tok)	36.0%	+4pp	1
Discrete “?” (2 tok)	48.0%	+16pp	1
Discrete mean	42.0%	+10pp	2
Random noise (2 tok)	51.6%	+19.6pp	10

Table 7: Discrete vs. continuous perturbation at 2 tokens. Continuous random noise outperforms the best discrete token by 3.6pp and the discrete mean by 9.6pp.

“/” (+4pp), indicating that even within the discrete regime, token identity matters—possibly reflecting different semantic or positional biases in the model’s embedding space.

4.10 Cross-Model Validation

To test whether the perturbation effect generalizes beyond Qwen3-4B, we run 2-token $n = 3$ scouts on three additional models using the same 25-task set and auto-calibrated RMS scale.

Model	Quant	Base	+Noise	Δ	Oracle	Rescued	Rescue%	Regress
Qwen3-4B ($n=10$)	4-bit	32%	51.6%	+19.6pp	100%	17/17	39%	0
Qwen3-8B ($n=10$)	8-bit	16%	28.8%	+12.8pp	80%	16/21	20%	0
DeepSeek-1.5B ($n=10$)	4-bit	76%	74.4%	−1.6pp	100%	6/6	—	0
phi-2 ($n=3$)	none	12%	18.7%	+6.7pp	28%	4/22	6%	0
Qwen3-1.7B ($n=3$)	4-bit	28%	29.3%	+1.3pp	44%	6/18	—	2
Qwen3-8B ($n=3$)	4-bit	24%	25.3%	+1.3pp	44%	7/19	—	2

Table 8: Cross-model 2-token perturbation results. Top: positive models (above midrule). Bottom: null results. *Rescued*: baseline failures solved by at least one direction (out of total baseline failures). *Rescue%*: per-direction rescue rate on baseline failures. *Regress*: baseline successes lost under all directions. The Qwen3-8B rows constitute a within-model quantization control (same architecture, tokenizer, task set; only quantization changed).

The effect replicates as a positive mean gain on three models: Qwen3-4B (+19.6pp), Qwen3-8B at 8-bit quantization (+12.8pp at $n=10$, oracle McNemar $p < 0.001$, 16 gains/0 losses), and phi-2 (+6.7pp, $p = 0.125$), an out-of-family model with no thinking mode. DeepSeek-R1-Distill-1.5B shows a distinct pattern: the mean across 10 random directions falls slightly below baseline (−1.6pp) but the oracle remains 100% (McNemar 6 gains, 0 losses, $p = 0.031$), indicating that every baseline failure is recoverable under at least one perturbation direction even when the average direction is unhelpful. This dissociation between

oracle accessibility and mean effect is consistent with DeepSeek’s high baseline (76%) leaving little room for uniform gain. The effect does not replicate on Qwen3-1.7B or Qwen3-8B at 4-bit quantization (both +1.3pp, oracle 44%, 2 regressions each).

Quantization $\times$ perturbation interaction. The Qwen3-8B 4-bit null is *explained* by an 8-bit adjudication experiment. Switching from 4-bit to 8-bit quantization on the same model with identical experimental parameters produces a crossover interaction: the baseline *drops* (24% $\rightarrow$ 16%) but the perturbation benefit *increases sharply* (+1.3pp $\rightarrow$ +12.8pp at $n=10$), oracle coverage rises (44% $\rightarrow$ 80%), and regressions vanish (2 $\rightarrow$ 0). Only 2 of 25 baseline tasks are shared between quantization levels; oracle sets overlap on only 9 of 25. This constitutes the cleanest within-model control in our study: architecture, tokenizer, task set, and perturbation budget are fixed; only weight precision varies. The mechanism is consistent with 4-bit quantization regularizing the trajectory landscape, helping the default reasoning path under a fixed decode budget but washing out small perturbation signals. 8-bit preserves a richer local trajectory structure, making perturbation-sensitive reasoning pathways accessible.

Substrate dependence. DeepSeek is *smaller* than both null Qwen models (1.5B vs. 1.7B and 8B at 4-bit) yet achieves 100% oracle coverage from $n=10$ directions. This rules out raw parameter count as the predictor. The oracle/mean dissociation on DeepSeek—every baseline failure is recoverable, but the average direction slightly hurts—suggests that reasoning distillation creates perturbation-sensitive trajectories that are direction-selective rather than uniformly beneficial. This points to training regime as a key substrate requirement, alongside sufficient quantization fidelity.

Cross-model dose-response. A dose-response on DeepSeek-R1-Distill-1.5B ($n=3$ per condition) confirms a non-monotonic constructive window: 1-token perturbation *hurts* (76% $\rightarrow$ 64%, -12 pp), 2-token yields the highest mean at $n=3$ (+5.3pp, SD = 0.046), and 3-token maintains a similar mean (+4.0pp) but enters a bifurcated regime (SD = 0.174, Cochran’s $Q = 9.5$, $p \approx 0.009$). At $n=10$, the 2-token mean drops to -1.6 pp while the oracle remains at 100% and Cochran’s Q becomes significant ($p = 0.025$), indicating that additional random directions enter the same heterogeneous regime observed at 3 tokens. This mirrors the Qwen3-4B pattern (1-tok marginal, 2-tok peak, 3-tok decline) in direction, establishing the non-monotonic window shape as a cross-model phenomenon, while the oracle/mean dissociation at $n=10$ demonstrates that high-baseline models may benefit primarily through task-selective direction search rather than uniform perturbation.

Task-level structure. One task (`nest_009`: $(47 \times 92 + 50) \bmod 28 = 6$) is unlocked by all tested models including Qwen3-1.7B, suggesting that modular arithmetic creates a “cheap compressed trajectory” that perturbation can discover regardless of model capacity. Cross-model difficulty analysis reveals a monotonic relationship: tasks that fewer models solve at baseline benefit *more* from perturbation (100% unlock rate for tasks failed by all models, vs. 0% for universally-solved tasks).

Statistical significance. Exact McNemar tests on paired baseline-vs-oracle task outcomes: Qwen3-4B $p < 0.0001$ (17 gains, 0 losses), Qwen3-8B 8-bit $p < 0.001$ (16/0 at $n=10$), DeepSeek $p = 0.031$ (6/0), phi-2 $p = 0.125$ (4/0). Fisher’s combined test across four positive models yields $p < 0.001$.

4.11 Cross-Domain Validation: Complex Planning Tasks

To test whether the perturbation effect extends beyond arithmetic, we run a 3-way comparison on five complex planning tasks: fraud detection system design, incident response planning, healthcare data platform architecture, Redis cache debugging, and Oracle-to-PostgreSQL database migration strategy. All conditions use Qwen3-4B (4-bit), greedy decoding (temperature = 0), and `max_new_tokens = 2048`.

Conditions. (1) Greedy baseline: standard generation, no modification (deterministic, single output). (2) Random perturbation: 2-token random embedding-scale noise, 5 seeds. (3) Evolution: trained latent scorer + evolutionary search $\rightarrow$ best evolved latent decoded via soft prompt projection, 5 seeds.

Finding 1: Attention sink avoidance. On the cache debugging task, greedy baseline produces only **14 words** before collapsing into degenerate output—a catastrophic generation failure. All five random perturbation seeds produce complete diagnostic plans of **650–710 words** with systematic investigation steps, root cause hypotheses, and resolution strategies. This binary rescue (complete failure → complete plan) demonstrates that soft prompt perturbation in the attention sink positions breaks degenerate greedy generation paths induced by attention concentration patterns [8].

Finding 2: Latent knowledge access via evolution. Evolved soft prompts do not merely produce more words—they access qualitatively different knowledge and reasoning paths. On the incident response task, evolution surfaces: honeypot deployment for attacker monitoring, MITRE ATT&CK framework analysis, tiered credential rotation with HSM integration, immutable container rebuilds from verified base images, and DMZ isolation of compromised services. None of these concepts appear in baseline or random perturbation outputs. On the database migration task, evolution converges to a structured recommendation with regulatory-audit-aware timeline, while baseline and perturbation outputs truncate before reaching a conclusion.

Caveats. The latent scorer is minimally trained. Evolution results are inconsistent across seeds—some seeds surface strong latents, others do not. An independent review ranked conditions as BASELINE > EVOLUTION > PERTURBATION on output “cleanliness” (baseline is shorter, less noisy). The quality advantage of perturbation and evolution is in completeness and knowledge diversity, not polish. These are five tasks with five seeds; the findings are suggestive, not statistically powered. Better scorers, evolution strategies, and aggregation methods should make the effect more consistent.

Implications. The planning results suggest that embedding-space perturbation operates through (at least) two mechanisms: (1) breaking degenerate greedy generation induced by attention sinks (achievable with random perturbation), and (2) steering attention toward different regions of the model’s knowledge space (achievable with directed evolution). The second mechanism represents a *new axis of improvement* orthogonal to scaling, fine-tuning, prompt engineering, retrieval augmentation, and sampling-based ensembles: it changes the continuous input representation to unlock latent knowledge with frozen weights. The efficiency advantage over best-of- N sampling is that evolution requires N forward passes through a small scorer MLP plus one generation pass, versus N full autoregressive generation passes.

5 Discussion

5.1 Two-Level Trajectory Model

The structural decomposition suggests a two-stage model:

1. **Magnitude sets the task-level threshold.** Perturbation energy determines which tasks are *unanimous* (solved by all directions), *frozen* (solved by none), or *sensitive* (solved by some). At $n = 10$, two tokens freeze zero tasks, three tokens freeze 5, and eight tokens freeze 2. The three-token frozen set is a strict superset of the two-token frozen set: over-perturbation contracts reachability.
2. **Direction selects within the sensitive set.** Each random direction solves a different subset of the sensitive tasks. Per-direction solve counts vary normally across all conditions (all $p > 0.3$ vs. heterogeneous iid null), so the diversity is in task identity, not in total solve count.

The two-token optimum emerges because it maximizes the reachable task set (25/25 at $n = 10$, vs. 20/25 for three tokens) while maintaining the highest mean per-direction accuracy ($12.9/25 = 51.6\%$).

5.2 Convergence vs. Computation

A grading-confound analysis reveals that the accuracy effect has model-dependent character. Extracting all integers from each response and checking whether the correct answer appears anywhere (not just as the last integer) shows:

Model	Grading Rule	Baseline	Perturbed	Δ
Qwen3-4B	Last integer (used)	32%	51.6%	+19.6pp
Qwen3-4B	Answer anywhere	80%	82%	+2pp
Qwen3-8B (8-bit)	Last integer (used)	16%	22%	+6pp
Qwen3-8B (8-bit)	Answer anywhere	32%	50%	+18pp

Table 9: Accuracy under two grading rules. Qwen3-4B can already compute the correct answer 80% of the time; its gain is purely convergence (placing the answer last). Qwen3-8B at 8-bit has a lower computational ceiling (32%); perturbation appears to improve computation as well as convergence. *Caveat*: answer-anywhere is computed from stored traces truncated at 2000 characters (>80% of responses are truncated), so these rates are lower bounds.

For Qwen3-4B, the model can already derive the correct answer in its reasoning chain for 80% of tasks but fails to produce it as the final output. Perturbation improves convergence without changing computational success. For Qwen3-8B at 8-bit, the model has substantial room for computational improvement (32% answer-anywhere at baseline), and perturbation produces apparent gains in both computation (+18pp) and convergence (+6pp), though these estimates are based on truncated stored traces and should be treated as preliminary. This pattern is consistent with the mechanism depending on the model’s computational ceiling, but confirming this requires full-trace analysis.

5.3 Trajectory Efficiency Examples

Response inspection confirms the convergence mechanism concretely:

- **nest_010** ($11 \times 21 \times 4 = 924$): The baseline response enters a verbose place-value explanation that exhausts the token budget before reaching the answer. Under noise, the model takes a more direct path ($11 \times 21 = 231$, $231 \times 4 = 924$) and completes within budget.
- **nest_006** ($= 6$): The baseline derives the correct answer within its reasoning trace but continues generating, runs out of tokens, and the last integer is not 6. Under perturbation, the model produces a clean answer ending with 6—a convergence fix, not a computation fix.

The effect is trajectory efficiency: noise routes computation through paths that terminate at the correct answer within the generation budget.

5.4 Why Two Tokens?

At $n = 10$ directions, two tokens have zero frozen tasks, compared to 5 at three tokens and 2 at eight tokens. At matched budget ($k = 3$), two tokens freeze 5.8 tasks on average versus 9.9 for three tokens and 7.9 for eight tokens. Two tokens thus maximize the reachable task set while maintaining the highest per-direction accuracy (mean $12.9/25 = 51.6\%$). We hypothesize that two tokens provide the minimum perturbation energy needed to productively diversify the reasoning trajectory while staying below the energy level that freezes marginal tasks. At one token, the perturbation is marginal; at three or more, the model’s reasoning trajectory diverges too early, exhausting the token budget before reaching an answer.

5.5 Implications for Practice

The training-free nature of this intervention suggests a practical “perturbation ensemble” strategy: run a small model with k random two-token prefixes and take the majority vote or oracle. Ten two-token directions achieve 100% coverage (25/25 tasks), while ten three-token directions reach only 80% and add no tasks beyond the two-token oracle. Beyond the two-token optimum, additional perturbation is deleterious: it lowers per-direction accuracy and contracts the reachable task set, freezing tasks that are solvable at two tokens.

6 Limitations

- **Model-contingent.** Cross-model scouts (Section 4.10) show positive transfer to four models (Qwen3-4B, Qwen3-8B at 8-bit, DeepSeek-1.5B, phi-2) but not to Qwen3-1.7B, suggesting the effect is substrate-dependent rather than universal. Full characterization across architectures and scales remains future work.
- **Cross-domain evidence is preliminary.** Planning task validation (Section 4.11) demonstrates qualitative effects (attention sink avoidance, latent knowledge access) on five tasks, but lacks the statistical power of the arithmetic study. A word problem scout ($n=3$, 25 tasks) yielded +2.7pp mean (not significant; McNemar 2/0, $p = 0.5$). The evolution findings depend on a barely-trained scorer and are inconsistent across seeds. Robust cross-domain generalization remains future work.
- **Small n .** 25 tasks with 10 perturbation directions at the key 2-token condition. Per-task mechanistic analyses remain statistically limited.
- **Quantization–perturbation entanglement.** The Qwen3-8B within-model control (Section 4.10) demonstrates that quantization modulates perturbation sensitivity: 4-bit yields a null result while 8-bit yields +16pp. This raises the question of whether our primary Qwen3-4B results are quantization-specific. Full-precision experiments are needed to determine whether 4-bit represents a sweet spot or whether higher fidelity would yield even larger effects.
- **Token budget interaction.** Regressions on easy tasks (large-answer category: 100% $\rightarrow$ 78%) and the 8-bit 8B baseline degradation (16% vs. 24% at 4-bit, with 8-bit traces being more verbose) suggest the effect interacts with the 1024-token generation budget. Perturbation may benefit models that can reach correct answers within budget, while harming those whose reasoning chains are already budget-limited.
- **Convergence vs. computation confound.** A grading-confound analysis (Table 9) reveals the model can already compute correct answers 80% of the time at baseline. For Qwen3-4B, the accuracy gain is primarily a convergence effect (placing the correct answer last). For Qwen3-8B at 8-bit, perturbation also improves computation (answer-anywhere 32% $\rightarrow$ 50%). The mechanism appears to depend on the model’s computational ceiling. DeepSeek shows the reverse: perturbation degrades both answer-anywhere (84% $\rightarrow$ 78%) and last-integer accuracy (76% $\rightarrow$ 69%), with only oracle coverage benefiting.

7 Conclusion

Random prefix perturbations at the embedding scale diversify reasoning trajectories in small language models, producing a sharp non-monotonic optimum. Two tokens raise Qwen3-4B mean accuracy from 32% to 51.6%. A grading-confound analysis reveals the mechanism is model-dependent: for Qwen3-4B (where the model already computes correct answers 80% of the time), the gain is primarily *convergence*—placing the correct answer as the final output. For Qwen3-8B at 8-bit (lower computational ceiling at 32% answer-anywhere), perturbation genuinely improves computation (+18pp answer-anywhere) as well as convergence.

The trajectory diversification is genuine and well-characterized: different random directions solve different task subsets, enabling 100% oracle coverage from 10 two-token directions on Qwen3-4B and 80% on Qwen3-8B at 8-bit. Per-direction solve counts vary normally; the diversity is in task identity, not total count. Cross-model validation confirms the effect on four models, while a within-model quantization control reveals that weight precision modulates perturbation sensitivity (4-bit null, 8-bit +12.8pp at $n=10$). A cross-model dose-response on DeepSeek confirms the non-monotonic window generalizes: 1-token hurts, 2-token is the local optimum, 3-token enters a bifurcated regime. At $n=10$, DeepSeek exhibits oracle/mean dissociation—every baseline failure is recoverable but the average direction slightly hurts.

Cross-domain validation on five complex planning tasks reveals two additional phenomena. First, perturbation rescues catastrophic attention-sink-induced generation failures: a 14-word baseline collapse becomes a 650+ word complete plan across all five perturbation seeds. Second, evolved soft prompts (via a trained latent scorer + evolutionary search) surface qualitatively different reasoning—security frameworks, architectural patterns, and diagnostic strategies absent from both baseline and random perturbation outputs—suggesting

that directed embedding-space perturbation can access latent knowledge the model possesses but cannot reach under standard decoding. This represents a new axis of improvement orthogonal to scaling, fine-tuning, prompting, retrieval, and sampling, with an efficiency advantage over best-of- N (one generation pass + N cheap scorer evaluations vs. N full generation passes). The planning findings are preliminary (five tasks, barely-trained scorer) but establish the mechanism’s relevance beyond arithmetic.

These findings establish embedding-space perturbation as a training-free intervention that diversifies reasoning trajectories and, when directed by evolution, can unlock latent knowledge inaccessible under standard decoding. Accuracy gains in arithmetic reflect a mixture of improved convergence and computation depending on the model’s baseline capability; in planning, the mechanism additionally breaks degenerate generation paths and steers attention toward different knowledge regions.

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